sustainability

MDPI

Article

Multi-Objective Path Planning for Unmanned Sweepers Considering Traffic Signals: A Reinforcement Learning-Enhanced NSGA-II Approach

Yiwen Huang [1], Wenjia Mou [1], Juncong Lan [1], Fuhai Luo [2], Kai Wu [1] and Shaofeng Lu [1,*]

[1] Shien-Ming Wu School of Intelligent Engineering, South China University of Technology, Guangzhou 511442, China; huangyiwen_@outlook.com (Y.H.); 202364820652@mail.scut.edu.cn (W.M.); juncong.lan@outlook.com (J.L.); whphwk@scut.edu.cn (K.W.);

[2] Fulongma Group Co., Ltd., 42 Longteng South Road, Xinluo District, Longyan 364000, China; fjlmlfh@163.com

* Correspondence: lushaofeng@scut.edu.cn

Abstract: With the widespread popularization of unmanned sweepers, path planning has been recognized as a key component affecting their total work efficiency. Conventional path planning methods often only aim to improve work efficiency while ignoring energy optimization, a crucial factor for sustainable development. In this paper, an energy- and time-minimization unmanned sweeper arc path problem (ETM-ARP) is investigated, and the effects of road slope, dynamic changes in on-board mass, mode switching of vehicle work, and traffic lights are taken into consideration to meet the requirements of a realistic structured road scenario. A new multi-objective mixed-integer nonlinear planning model is proposed for this problem. To solve this model, we propose a deep Q-network (DQN) and Adaptive Large Neighborhood Search Algorithm (ALNS)-driven non-dominated sorting genetic algorithm II (QALNS-NSGA-II). The novelty of this algorithm lies in integrating DQN into ALNS, to guide high-quality adaptive operator selection during the search process based on additional information. The computational results of various examples confirm the effectiveness of the proposed method. The proposed method can be used to improve the efficiency and sustainability of unmanned sweepers for sweeping on structured roads.

Keywords: unmanned sweeper; path planning; multi-objective optimization; DQN

check for updates

Citation: Huang, Y.; Mou, W.; Lan, J.; Luo, F.; Wu, K.; Lu, S. Multi-Objective Path Planning for Unmanned Sweepers Considering Traffic Signals: A Reinforcement Learning-Enhanced NSGA-II Approach. *Sustainability* **2024**, *16*, 11297. https://doi.org/10.3390/su162411297

Academic Editors: Angelo Coppola and Gennaro Nicola Bifulco

Received: 20 November 2024
Revised: 14 December 2024
Accepted: 19 December 2024
Published: 23 December 2024

Correction Statement: This article has been republished with a minor change. The change does not affect the scientific content of the article and further details are available within the backmatter of the website version of this article.

Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

With the rapid development of the economy, the capacity of urban and rural environmental construction has been significantly improved, and the sanitation automobile industry has gradually transformed to electrification and intelligence. Unmanned sweepers have been widely promoted in recent years due to their advantages of high efficiency and low cost [1], and are expected to become the main force of urban street cleaning in the future. Route planning is the key to unmanned sweeper operation, directly affecting work efficiency. Unmanned sweepers need to start from a designated location to cover a specific neighborhood, during which they may need to stop and wait at signalized intersections and pass quickly over repeated passing sections to improve efficiency. The core of the solution to unmanned sweeper path planning lies in determining the most cost-effective full-coverage routes, often optimizing travel distance, energy consumption, or time to improve sweeping efficiency while satisfying constraints, which supports the sustainability of smart cities.

For structured urban roads, traffic signals are common traffic control facilities, and vehicles often encounter multiple signal-controlled intersections in the course of traveling, which may cause interruptions to vehicle movement. At present, in the path planning algorithms that consider the influence of traffic signals, scholars at home and abroad have proposed algorithm models from different perspectives. Song et al. [2] proposed a path

planning model based on traffic signal state and traffic condition perception to dynamically plan the shortest path for a vehicle in real time using an A2C-based deep reinforcement learning method. Experiments showed that compared with the existing solutions, the proposed method has a significant effect, reducing driving time and waiting time. Karoui et al. [3] proposed a Green Light Optimal Speed Advisory (GLOSA) system based on path planning. Using the Dijkstra algorithm in large-scale scenarios, they evaluated the function of GLOSA under free-flow and peak-hour traffic conditions, reducing fuel consumption by an average of 11% and up to 47.4%. However, these studies did not consider in detail the impact of different driving styles on energy consumption when vehicles pass through signalized intersections. In fact, idling, braking, and acceleration behaviors caused by traffic signals are the main contributors to time wastage and energy consumption [4]. Georgiadis et al. [5] incorporated simulated environmental factors such as traffic lights and stop signs in their model and accurately estimated the energy consumption of electric vehicles. The results showed that the path planning model managed to reduce energy consumption by about 13% although it increased traveling time by 6%.

In the application scenario of unmanned sweepers, the vehicle serves road segments rather than customer nodes on the road. This type of path planning problem is known as the Arc Routing Problem (ARP). For the optimization of ARP problems, different objective functions have been proposed in the literature, mainly including minimizing travel distance [6–8], minimizing travel time [9,10], minimizing total cost (e.g., vehicle configuration cost and operation cost, etc.) [11,12], and minimizing vehicle energy consumption [13,14]. Route selection is a major factor influencing vehicle energy consumption. However, some articles tend to overlook factors that affect the energy consumption of electric vehicles [14,15], such as changes in load mass, speed, road slope, and regenerative braking systems. Instead, they simplify the relationship between a vehicle's energy consumption and the distance traveled to a linear one. Boriboonsomsin et al. [16] showed that there is a conflict between the shortest distance, shortest time, and most energy efficient paths, where the most energy-efficient path reduces the energy consumption by 12% to 14%, but extends the travel time by 16% to 22% compared to the shortest-distance path. Based on this, this study proposes a multi-objective unmanned sweeper arc path problem (ETM-ARP) with the objective of minimizing energy consumption and travel time, aiming to analyze and provide trade-offs between the two. The optimization of this problem is important for improving the cleaning efficiency of unmanned sweepers and promoting environmental sustainability.

The ARP was first proposed by Golden et al. [17] and has been proven to be an NP-hard problem. Since ETM-ARP involves various constraints (e.g., signal light constraints, full road coverage constraints) and nonlinearities (e.g., vehicle energy consumption), finding an optimal solution in a reasonable time is a challenging task. Several approaches have provided potential solutions in recent years, mainly including exact and metaheuristic algorithms. The exact algorithms mainly include the cutting plane [11], the column generation method [18], and branch-and-bound [19]. However, this kind of method makes it difficult to obtain a good feasible solution for large-scale problems in a short time. Metaheuristic algorithms include ant colony algorithms [20], genetic algorithms [21], the Adaptive Large Neighborhood Search Algorithm [22], etc. However, heuristic algorithms suffer from suboptimality, lack of convergence, and limited generalization when dealing with real-life optimization problems.

Reinforcement learning (RL), a subfield of machine learning, learns to make optimal decisions by interacting with the environment. RL agents maximize long-term payoffs by mapping the current state to actions and continually optimizing the decision-making strategy through a trial-and-error process [23]. In recent years, more and more studies have combined reinforcement learning with metaheuristics to form a metaheuristic approach for extracting knowledge from data, and this approach improves algorithm performance by dynamically adjusting the decision-making strategy during the search process [24–26].

Therefore, in this paper, deep reinforcement learning is integrated into the adaptive layer of ALNS for tuning key parameters in the algorithm to solve ETM-ARP.

Due to the need for the sustainable development of smart cities, pure cleaning efficiency can no longer meet the needs of unmanned sweeper path planning. Therefore, in this paper, we propose a multi-objective path planning model for unmanned sweepers (ETM-ARP), aiming to improve cleaning efficiency and reduce energy costs. In order to solve the shortcomings of traditional heuristic algorithms relying on experience and problems, this paper proposes a DQN-driven hybrid metaheuristic algorithm (QALNS-NSGA-II), which effectively alleviates the problem that traditional heuristic algorithms are prone to falling into local optimal solutions by introducing DQNs to learn the optimal behaviors of the heuristic operators. The main contributions of this paper are summarized as follows:

- We propose a new multi-objective mixed-integer nonlinear planning model (ETM-ARP) that aims to minimize the energy consumption (E) and travel time (T) of the unmanned sweeper by taking into account factors such as traffic signals and work mode switching.
- For the proposed ETM-ARP problem, we incorporate DQN into ALNS for adaptively selecting operator choices and parameter configurations in ALNS during the search period, and design destruction and repair operators.
- The proposed QALNS is embedded into non-dominated sorting genetic algorithm II (NSGA-II), and specialized reward functions and acceptance levels are designed for the movement of Pareto frontier solutions to obtain higher diversity and quality. The experimental results demonstrate that our algorithm performs superiorly in terms of the computational efficiency, quality, and diversity of Pareto frontier solutions.

The remainder of the paper is organized as follows. Section 2 introduces the unmanned sweeper energy consumption model and traffic light model, and establishes the ETM-ARP. A detailed QALNS-driven NSGA-II (QALNS-NSGA-II)-based approach is presented in Section 3, which includes the design of the operators and reinforcement learning framework. Then, test experiments are conducted in Section 4, and the results are compared and analyzed. Section 5 gives a summary of this paper and proposes topics for future work.

2. Mathematical Modeling

This section describes the ETM-ARP problem established in this paper. First, the directed graph and routing-related constraints are defined. Then, a traffic light model of the road network and a detailed energy consumption model of the unmanned sweeper are presented, in which various constraints are considered, such as vehicle dynamics, potential energy difference, and battery efficiency. Finally, a multi-objective path planning model that allows the unmanned sweeper to minimize its energy consumption (E) and travel time (T) is developed.

2.1. Problem Description

Consider a complete directed graph $G = (V, E)$, where V denotes the set of n vertices in the graph, and the arc set $E = \{e_1, e_2, \cdots, e_m\}$ denotes that there are m arc segments in the graph, which includes the demand set R_w and the non-demand set R_n. The unmanned sweeper vehicle sweeps the road network by traversing all the demand sets R_w to complete the road sweeping work. The following assumptions are made in this paper:

- The distance between locations is measured using the Euclidean distance.
- The unmanned sweeper is fully charged at the start of the sweeping task and does not need to be recharged during the sweeping route.
- The unmanned sweeper has two modes, sweeping mode and cruising mode, corresponding to different speeds, v_{clean} and v_{cruise}.
- The energy consumption model of the unmanned sweeper is a nonlinear model that takes into account the longitudinal dynamics of the vehicle.
- The impact of other vehicles, non-motorized vehicles, and pedestrians on unmanned sweepers is not taken into account.

- From the point of view of safe passage and model simplification, a yellow light is categorized as a red light state, and the unmanned sweeper is set to stop in time when it encounters a yellow light.

2.2. Unmanned Sweeper Energy Consumption Model

During driving, the vehicle's propulsion system must generate sufficient driving force to overcome the resistance forces encountered to propel the vehicle forward, which includes air resistance, rolling resistance, slope resistance, and acceleration resistance, so it is necessary to consider the longitudinal dynamic equation of the vehicle, which is shown below:

$$M\frac{dv}{dt} = F_{\mathrm{w}} - Mgsin\theta - Mgc_r cos\theta - \frac{1}{2}\rho_a A c_d v^2 \tag{1}$$

where ρ_a is the air density, A is the frontal surface area, c_d is the aerodynamic drag coefficient, M is the whole vehicle mass of the sweeper, g is the gravitational acceleration, c_r is the rolling resistance coefficient, θ is the angle of the road slope, v is the traveling speed of the sweeper, and F_w is the traction or braking force.

Considering the conversion efficiency eff_m of the vehicle motor and the charging/discharging efficiency eff_b of the battery, the mechanical energy consumed by the vehicle is converted into electrical energy in the battery, and the relationship between energy conversion traction/regenerative braking is as follows:

$$E_{ij} = \begin{cases} \dfrac{F_w \cdot \nu \cdot \Delta t}{eff_m^d \cdot eff_b^d} & F_w \geq 0 \\ F_w \cdot \nu \cdot \Delta t \cdot eff_m^r \cdot eff_b^r & F_w < 0 \end{cases} \tag{2}$$

where E_{ij} denotes the energy consumption (+) or regenerative braking energy ($-$) on the (i,j) road segment, eff_m^d and eff_m^r are the motor efficiency of the vehicle motor during traction and regenerative braking, respectively, and eff_b^d and eff_b^r are the discharge/charge efficiency of the battery, respectively.

2.3. Traffic Signal Model

Define the traffic signal information set as $\mathcal{S} = \{\mathcal{S}_n | n \in \mathcal{J}\}(\mathcal{J} = \{1, 2, \cdots, N_s\})$, where each subset includes the nth signal position and SPaT information, $\mathcal{S}_n = \left\{ l^n, T_{\mathrm{g}}^n, T_{\mathrm{r}}^n, t_\phi^n \right\}$, where l_n is the signal position, T_{g}^n and T_{r}^n are the green light and red light timing lengths, respectively, and t_ϕ^n is the initial phase time of the nth signal light. It is noted that from the point of view of safe passage, it is not recommended for vehicles to pass through the intersection in the yellow light state, so in this paper, a yellow light is categorized as a red light, and we only consider the two states of green light and red light signals.

Assume that a complete signal cycle is red–green. Therefore, the standard cycle length of a signal is defined as $T_{\mathrm{l}}^n = T_{\mathrm{g}}^n + T_{\mathrm{r}}^n$; considering the difference in cycle lengths of each signal, the number of cycles of the nth signal at moment t is calculated as

$$C^n = \lceil \frac{t - (t_\phi^n + T_{\mathrm{g}}^n)}{T_l^n} \rceil + 1 \tag{3}$$

where $\lceil \cdot \rceil$ is the integer up function.

Therefore, the start time t_{g,C^n} and end time t_{r,C^n+1} of the green light of the nth signal in the C^n cycle can be calculated as

$$t_{g,C^n} = t_\phi^n + T_r^n(C^n - 1) + T_g^n(C^n - 1) \tag{4}$$

$$t_{r,C^n+1} = t_\phi^n + T_r^n(C^n - 1) + T_g^n C^n \tag{5}$$

It can be obtained that the passable green window $\mathbb{C}_t^n$ of the nth signalized intersection at time t is

$$\mathbb{C}_t^n = \left[t_{g,C^n}, t_{r,C^n+1}\right] \tag{6}$$

Therefore the passage time at the nth signalized intersection at time t can be obtained as follows:

$$t_{wait} = \begin{cases} 0 & t \in \mathbb{C}_t^n \\ t_{g,C^n} - t & else \end{cases} \tag{7}$$

2.4. Problem Formulation of ETM-ARP

The objective considered in this model is to determine a set of routes for the unmanned sweeper, and in order to achieve efficient and sustainable cleaning operations, we construct a multi-objective optimization model (ETM-ARP) to minimize the T of unmanned sweepers and to reduce the E in order to reduce the environmental impact. Based on the above description, the objective function can be expressed as follows:

$$f = \min(E, T) \tag{8}$$

$$\mathrm{E} = \sum_{k=1}^{K} EC(v_k, M_k, d_k) \tag{9}$$

$$\mathrm{T} = \sum_{k=1}^{K} \left(\frac{d_k}{v_k} + t_{wait}\right) \tag{10}$$

The constraint functions are as follows:

$$\sum_{k=1}^{K} \alpha_{i,j,k} = 1, \forall (i,j) \in R_w \tag{11}$$

$$\alpha_{i,j,k} + \beta_{i,j,k} \leq 1, \forall k, \forall (i,j) \in E \tag{12}$$

$$\sum_{j \in \delta^+} \left(\alpha_{i,j,k} + \beta_{i,j,k}\right) - \sum_{j \in \delta^-} \left(\alpha_{j,i,k+1} + \beta_{j,i,k+1}\right) = 0, \forall k, \forall i \in N \setminus \{o, e\} \tag{13}$$

$$u_{i,k} - u_{j,k} + M\left(\alpha_{j,i,k} + \beta_{j,i,k}\right) \leq M - 1, \forall k, \forall (i,j) \in E \tag{14}$$

$$v_k = v_{clean}\alpha_{i,j,k} + v_{cruise}\beta_{i,j,k}, \forall k, \forall (i,j) \in E \tag{15}$$

$$M_{k+1} = M_k + m_{i,j}\alpha_{i,j,k}, \forall k, \forall (i,j) \in E \tag{16}$$

$$\alpha_{i,j,k} \in \{0,1\}, \forall k, \forall (i,j) \in E \tag{17}$$

$$\beta_{i,j,k} \in \{0,1\}, \forall k, \forall (i,j) \in E \tag{18}$$

Equations (8)–(10) are the minimization objective functions, i.e., minimizing time and energy consumption. $\alpha_{i,j,k}$ and $\beta_{i,j,k}$ denote whether the unmanned sweeper adopts the sweeping mode or the traveling mode when it passes through the road segment (i,j) for the kth time, respectively. δ^+ and δ^- denote the set of nodes that depart from and arrive at the node, respectively. Equation (11) ensures that each section of a given road must be swept once by the sweeper. Equation (12) ensures that the unmanned sweeper chooses either the sweeping mode or the cruising mode when traveling on any road segment. Equation (13) is a flow balance constraint. Equation (14) is a sub-loop prevention constraint. Equation (15) ensures that the speed of the unmanned sweeper on the current roadway changes with the mode selection. Equation (16) represents the total mass of the unmanned sweeper, consisting of the vehicle mass and the dynamic load. The total mass M of the unmanned sweeper vehicle rises cumulatively as its sweeping work proceeds. At the beginning of the path, the total mass of the unmanned sweeper is equal to the vehicle mass.

3. Improved Multi-Objective Planning Algorithm Based on Reinforcement Learning

The model presented in Section 2 is a multi-objective mixed-integer programming model. In multi-objective optimization problems (MOPs), a unique global optimal solution typically does not exist. Instead, the optimization results are often represented in the form of a Pareto optimal solution set, consisting of non-dominated solutions. This paper aims to find the Pareto-optimal solution set of the proposed model in order to balance the trade-offs among the objectives and provide multiple solutions for unmanned sweeper path planning.

In this section, we propose a multi-objective planning algorithm based on improved deep reinforcement learning (QALNS-NSGA-II). Figure 1 shows an algorithmic framework diagram of QALNS-NSGA-II. In Step 1, we employ a method combining random generation and opposition-based learning (OBL) to produce initial feasible solutions. This approach, which simultaneously considers initial guesses and their opposites, has been demonstrated to perform well in [27]. In step 2, an elite preservation strategy is used for retaining the optimal individuals [28]. This process utilizes fast non-dominated sorting and crowding distance calculation. In step 3 a modified ALNS algorithm based on DQN (QALNS) is used for generating new populations; then QALNS is utilized to drive NSGA-II until the convergence criteria are met. Finally, the final obtained Pareto frontier solution is outputted in step 4.

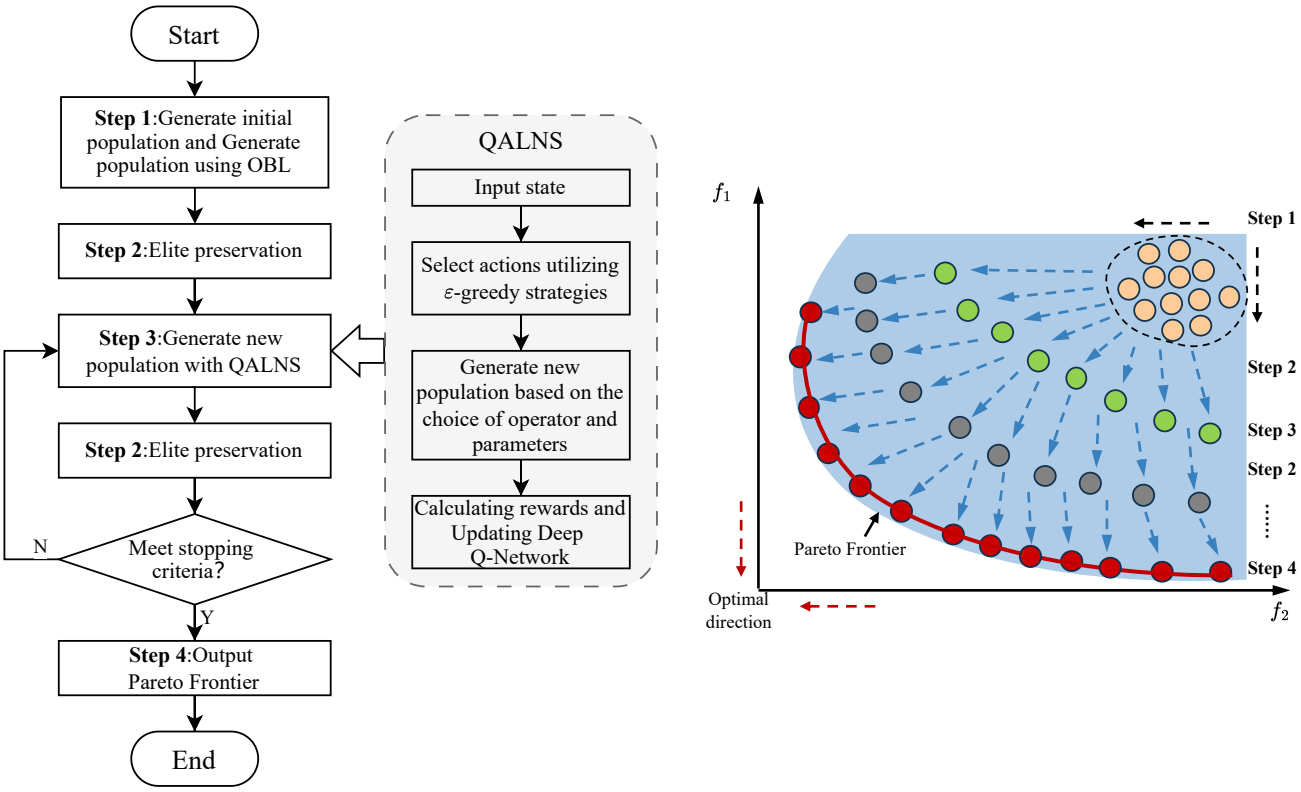


Figure 1. (**a**) A flowchart of the QALNS-NSGA-II algorithm. (**b**) An illustration of the execution of the QALNS-NSGA-II algorithm, where the step in (**b**) corresponds to the step in (**a**).

The following is a detailed description of QALNS in Step 3: First, the encoding and decoding methods are designed for the ETM-ARP problem; then the corresponding ALNS operators are designed; finally, DQN is used to improve the adaptive layer of the ALNS algorithm.

3.1. Encoding and Decoding

Encoding aims to transform the complete planning path sequence into a chromosome. Considering the goal of ETM-ARP to accomplish full road sweeping coverage, in this paper, we adopt a coding scheme that numbers all road segments to be swept and sequences the numbered task segments.

Decoding is the inverse process of encoding, which transforms chromosomes into solutions. In this paper, a greedy decoding scheme is designed by considering the shortest path insertion and pruning to achieve decoding and ensure the total path length is minimized under the current task order. The specific decoding method is shown in Algorithm 1.

Algorithm 1 Decoding Process

Input: Service Arc Sequence Sol,Neighborhood matrix of arc segments SP
Output: The generated path $Path$

1: **STEP 1: Route generation**
2: **while** $i < len(Sol) - 1$ **do**
3: $j = i + 1$
4: Get the shortest path $InsertPath$ connecting arcs $Sol[i]$ and $Sol[j]$ by looking up the adjacency matrix table and inserting it into the sequence of arcs.
5: **if** $len(InsertPath) \geq 1$ **then**
6: Iterate through the successor sequence of $Sol[j]$ to check if any arc segments are contained in $InsertPath$, and delete if present
7: **end if**
8: **end while**
9: **STEP 2: Route pruning**
10: Get the initial path $path_{ini}$ in STEP 1
11: Iterate over the initial path $path_{ini}$, remove the head node if it appears in subsequent paths
12: Loop the steps 11 until there is no repeating of the current head node, eventually getting the path $path$

3.2. Removal and Repair Operator

When dealing with the initial solution and the solutions generated during subsequent iterations, different operators need to be set up for neighborhood operations according to specific optimization needs. These operators are divided into two groups: removal operators and repair operators. The removal operator destroys the structure of the solution, while the repair operator repairs the solution to obtain a new one. By reasonably combining the operators and adjusting the parameters, the performance of the algorithm and the solving ability can be effectively improved.

3.2.1. Removal Operator

Random removal: During the neighborhood search, N non-repeating arc segment numbers are randomly selected as removal objects to destroy the original sequence.

Energy Greedy Removal: For this problem, the saving space of E mainly lies in the loss of additional loads; therefore, the additional energy consumption ratio is defined as $rate = \frac{E_{extra}}{|E_{vehicle}|}$, where E_{extra} is the energy consumption of the additional loads and $E_{vehicle}$ is the energy consumption of the vehicle body itself. During the neighborhood search, the growth rate of additional energy consumption of the current arc segment is calculated ($growth = \frac{rate_n}{rate_{n-1}}$), and the largest N sections are selected for removal.

Time Greedy Removal: During the neighborhood search, the signal light waiting time for the current arc segment is calculated t_{wait}, and the N segment with the largest waiting time is selected for removal.

Beam Removal: During the neighborhood search, the current solution is divided into N subsets that satisfy $t_{wait} = 0$. The subset of subsets that contains the maximum number of service arc segments is selected for removal.

3.2.2. Repair Operator

Random Repair: During the neighborhood search, we randomly select N non-repeating insertion positions and insert the removed objects into them.

Greedy Repair: During the neighborhood search, we insert the removed object in front of the one that has a higher growth rate of additional energy consumption than the currently removed objects.

Beam Repair: During the neighborhood search, all positions are traversed sequentially, all removed objects are inserted uniformly, and the position with the smallest time is selected for insertion.

3.3. RL-Based ALNS Strategy Selection

Due to the lack of effective knowledge accumulation, the traditional ALNS method is prone to local optimization and slow convergence. To improve this deficiency, we introduce deep reinforcement learning to drive ALNS through DQN in the iterative process to dynamically select the most suitable operation operators and parameters, to enhance the convergence and global search capability of the algorithm. In this section, the basic elements of DQN-based ALNS strategy selection are defined, including search space, agent, state, action, and reward. In this paper, each solution is considered as an agent, and the search space can be regarded as the environment in which the agent is located. The actions, states, and rewards of the agent are defined as follows.

3.3.1. State

In order to obtain high-quality and diverse Pareto frontier solutions, it is required to design the state of reinforcement learning with representations of the features that react to its performance during the search process, thus guiding the agent to choose the best action (operator) during each iteration. Therefore, in this paper, we design the agent's state as follows:

$Energy_{cur}$ and t_{cur} denote the E and T of the current solution, respectively; $Energy_{min}$ and t_{min} denote the minimum E and the minimum T, respectively; and $iteration$ denotes the current number of iterations. These five state characteristics enable the agent to know roughly how it performs during the search period, and help the agent to select an effective strategy persistently throughout the search process to assist at a macro level.

$Energy_{gap}$ and t_{gap} denote the difference in E and T between the current solution and the optimal solution, respectively, and $accept$ indicates whether the current solution is selected or not. These states help the agent balance the diversity of Pareto frontier solutions and the speed of convergence in the current step.

$was_changed$ denotes whether the current solution is different from the previous one, $unseen$ indicates whether the current solution has been encountered during the search, $last_action$ indicates the operation selected in the previous step, and $improved$ indicates whether the current solution has been improved compared to the previous one. These states provide information about the results of the agent's operations in the previous step and the current one, and help to avoid ineffective repetition of choices at a micro level.

3.3.2. Reward

The reward function is a measure of the goodness of the action in a given state. Based on the selected state information and training goal expectation, the total reward function is set as follows:

$$R_{total} = R_1 + R_2 \tag{19}$$

where the reward function R_1 is used to evaluate the performance of the current operation, and the corresponding rewards are given sequentially according to the quality of the new

solutions found by the agent each time, which can motivate the agent to use diversified operators, and the reward setting is referred to the scoring mechanism in ALNS, as shown in (20):

$$R_1 = \begin{cases} 5, & \text{if } pop \prec p \\ 3, & \text{if } Energy_{pop} < Energy_p \text{ or } t_{pop} < t_p \\ 1, & \text{if accpet}(pop) \\ 0 & else \end{cases} \tag{20}$$

where $\prec$ denotes domination, which means that the current solution is not inferior to another solution on all objectives and is better than another solution on at least one objective.

The reward function R_2 considers the quality performance of the current solution in the solution set, as shown in (21):

$$R_2 = \begin{cases} 5, & \text{if } rank = 1 \\ 0, & else \end{cases} \tag{21}$$

where *rank* denotes the non-dominated rank of the solution in the solution set.

3.3.3. Action

For each iteration, the agent will take different actions based on different states, and the set of actions contains 36 actions in total, including the choice of removal operators (4) and repair operators (3), and the number of operations (3).

The action selection strategy is the method by which the agent searches through the action set, and in this paper, we use the ε-greedy strategy, as shown in (22). ε decays throughout the iteration, making the agent more likely to choose random actions to explore the search space at the beginning of the iteration and to choose the best action to enhance utilization based on the learned experience at the later stages of the iteration.

$$A(s) = \begin{cases} \arg\max_a Q(s, a') & \text{with probability } \varepsilon \\ randomly & \text{with probability } 1 - \varepsilon \end{cases} \tag{22}$$

4. Numerical Experiments

In this section, the performance of the proposed model and algorithm is evaluated in different cases. First, performance indicators are defined to measure the performance of the algorithms. Due to the lack of standardized benchmark test cases, the test cases are generated according to certain rules. Then, the importance of including characteristics such as dynamic changes in on-board mass and slope in the energy function is evaluated. Third, the QALNS-NSGA-II method proposed in this paper is compared with ALNS-NSGA-II and classical NSGA-II. Finally, sensitivity analysis experiments are performed.

The numerical experiments used PyCharm Community 2023.3 on a PC with a 12th Gen Intel(R) Core(TM) i7-12700F 2.10GHz processor and 16 GB RAM.

4.1. Experimental Settings

To the best of our knowledge, there are limited cases of the ETM-ARP problem, so we generate test cases as described below. The mass of road waste in each section follows a truncated normal distribution $G \sim \max(0, N(3, 8^2))$ in kg. The elevation of each vertex was randomly generated between 0 and 100 m, and the average slope was calculated from the difference in elevation and distance between the two vertices. The slopes were calculated to be limited in the range of $[-0.1, 0.1]$. The unmanned sweeper used in this case was the Snail White S100-N unmanned sweeper, the specific parameters were determined based on the product information on the official website [29], and the setting values are shown in Table 1:

Table 1. Unmanned sweeper specification parameters.

Parameter	Description	Value
$m_{vehicle}$	Unmanned sweeper mass	395 kg
v_{clean}	Driving speed (clean)	5 km/h
v_{cruise}	Driving speed (cruise)	20 km/h
g	Gravitational acceleration	9.81 m/s^2
A	Frontal surface area	1.1 m^2
ρ	Air density	1.2041 kg/m^3
c_r	Coefficient rolling resistance	0.01
c_d	Coefficient aerodynamic drag	0.7
eff_m^d	Motor drive efficiency	85%
eff_m^r	Motor regenerative braking efficiency	84.7%
eff_b^d	Battery discharge efficiency	93%
eff_b^r	Battery charging efficiency	90%

The road networks for the test cases were obtained from the CARP benchmark cases egl [30], val [31], gdb [32], kshs [33], and Sioux Falls [34], and the datasets egl, val, gdb, and kshs are publicly available at http://www.uv.es/ belengue/carp.html (accessed on 18 September 2024). Each test case is written as "Case name-number of nodes-number of arcs". For example, a test case using the road network of the city of Sioux Falls, which has 24 nodes and 76 arc segments, is denoted as "SiouxFalls-24-76". To evaluate the effectiveness of the proposed methodology, we consider 10 different instances and run each instance five times to obtain the mean (mean) and standard deviation (std) used for the evaluation.

Second, performance indicators are defined to measure the performance of the algorithm. The quality of MOP is judged based on the convergence and diversity of the results, where convergence reflects the distance between the obtained Pareto frontier solution and the optimal Pareto solution, and diversity reflects whether the distribution of Pareto frontier solutions is uniform. Here, we use the following indicators to compare these three aspects:

(1) Coverage Metric (CM) [35]: This indicator represents the proportion of solutions in the non-dominated solution set B that are dominated by the non-dominated solution set A. The closer the CM value is to 1, the better set A is; the CM is calculated using (23):

$$\mathrm{CM}(A,B) = \frac{|\{b \in B | \exists a \in A : a \prec b\}|}{|B|} \tag{23}$$

CM(A,B) denotes the dominance of set A over set B. If every solution in set B finds a corresponding solution dominating in set A, then CM(A,B) = 1. If CM(A,B) > CM(B,A), then it means that set A is of better quality than set B.

(2) Hypervolume (HV) [36]: HV represents the region enclosed by the set of all non-dominated solutions obtained by the algorithm and the reference points in the target space, and measures comprehensive performance based on convergence and diversity; a higher value of HV indicates better performance. HV is defined as

$$HV = \lambda \left(\bigcup_{i=1}^{|A|} v_i \right) \tag{24}$$

where λ denotes the Lebesgue measure, v_i denotes the hypervolume surrounded by reference points and non-dominated individuals i, and A denotes the set of non-dominated sets.

(3) Spacing Matrix (SM) [37]: SM denotes the standard deviation of the minimum distance between each solution and other solutions in the Pareto frontier solution, which reflects the uniformity of the distribution of individuals in the set of non-dominated solutions,

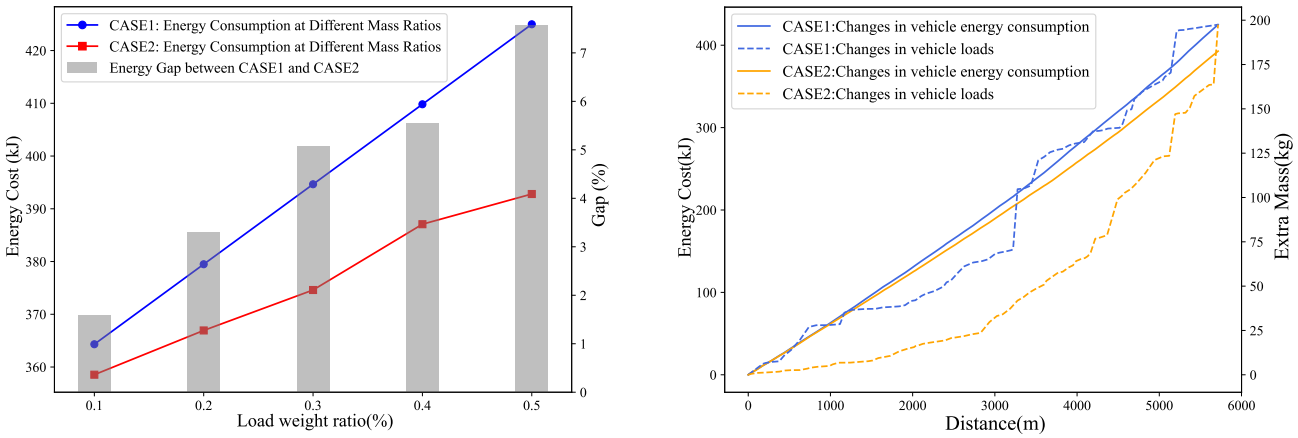
and is used to evaluate the diversity of the Pareto frontier. The smaller the SM is, the more uniform the distribution of the solutions obtained, and the better the diversity of the population. SM is calculated as

$$\mathrm{SM} = \sqrt{\frac{1}{|A|-1}\sum_{i=1}^{|A|}\left(\overline{d}-d_i\right)^2} \tag{25}$$

where A is the set of non-dominated solutions, d_i represents the minimum distance between the ith solution and any other solution in the non-dominated set A, $d_i = \min_{j\in A, i\neq j}\left(\sum_{k=1}^{M}\left|f_k(i)-f_k(j)\right|\right)$, and $f_k(\cdot)$ denotes the kth target value of the individual. $\overline{d}$ is the mean value of the distance between individuals in the solution set, which can be expressed as $\overline{d} = \frac{\sum_{i=1}^{|A|} d_i}{|A|}$.

4.2. Evaluating the Importance of Including Dynamic Loads and Road Slope in the Energy Function

To emphasize the importance of including dynamic loads and road slope in the energy consumption model, we consider two different cases designed for comparative analysis using the road network of Sioux Falls, which consists of 24 nodes and 76 arcs. In the first case, we assume that the dynamic changes in onboard mass and road slope are ignored, and set the parameters $\Delta m = 0$ and $\Delta\theta = 0$; in the second case, we take into account the influence of the dynamic changes in on-board mass and road slope. To ensure the stability and range of the unmanned sweeper, the onboard mass usually accounts for 10–50% of the vehicle's mass. In this experiment, we set the load to 10–50% of the vehicle mass for testing to comprehensively examine the difference in E between the two cases under different load ratios. The results are shown in Figure 2a:



(**a**) (**b**)

Figure 2. (**a**) Energy consumption (E) at different load weights for two cases. (**b**) Comparison of energy consumption (E) and quality changes in planning results of two cases.

As can be seen from Figure 2a, the total E shows a significant upward trend as the load share increases. However, the rise in E is relatively flat when the effects of dynamic load and slope are considered. As the load ratio increases, the gap between the E of the two cases gradually widens, indicating that the energy-saving effect of the optimization strategy is more significant in the high-load case.

For a load weight ratio of 50%, the optimal E results for the unmanned sweeper in the two cases are shown in Table 2 and Figure 2b.

Table 2. Planning results in two cases.

Case	Distance (m)	Repeat (%)	E (kJ)	
			Value	Mean
CASE 1: Ignore dynamic mass and slope	7937.5	100	424.984	5.592
CASE 2: Consider dynamic mass and slope	7937.5	100	392.801	5.168

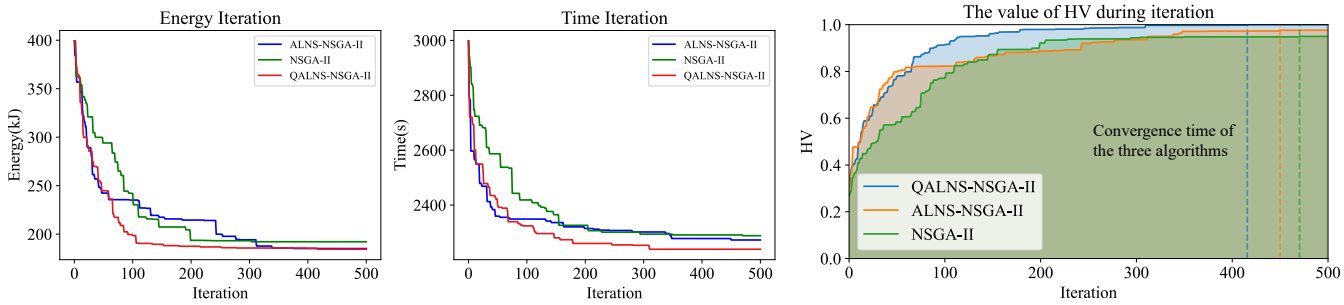
By comparing these two cases, it can be seen that considering the effects of dynamic loads and road slope can achieve nearly a 7.6% reduction in E for unmanned sweepers without affecting the path length. In CASE 1, the idealization of the relationship between vehicle E and path length without considering the effects of road slope and dynamic loads leads to underestimation of the actual E of the vehicle, especially in the case of high on-board mass or complex slopes. In CASE 2, when considering the effects of dynamic load and road slope, path planning can allocate energy more reasonably according to the load and slope. As can be seen from the orange dotted line in Figure 2b, the unmanned sweeper tends to choose a light load at the beginning and a heavy load at the later stage, so as to minimize unnecessary energy waste. Considering realistic factors such as dynamic load and road slope can make the model closer to real-world conditions, which is conducive to the rational use of the environment and reducing unnecessary E when the unmanned sweeper is applied in practice.

4.3. Algorithm Comparison and Analysis

In this section, a series of computational experiments are performed to validate the proposed algorithm, comparing QALNS-NSGA-II with ALNS-NSGA-II and classical NSGA-II.

4.3.1. Convergence Rate Evaluation

Figure 3a shows the convergence curves of the three algorithms under the objective of E and the objective of T. The comparison shows that the proposed QALNS-NSGA-II has better convergence speed than the other algorithms in exploring the boundary of the Pareto frontier solution for both E and T. In terms of the degree of convergence, QALNS-NSGA-II is less likely to fall into local optimal solutions compared to other algorithms, which implies that the algorithm has stronger global search capability, which is conducive to exploring more cutting-edge Pareto frontier solutions.



(**a**) (**b**)

Figure 3. (**a**) Convergence curves of the val-24-78 case. (**b**) The change in hypervolume for the val-24-78 case.

However, evaluating the convergence of the Pareto frontier solutions not only depends on the speed of exploration of the boundary solutions, but the convergence of the overall solution set should also be considered. In MOP, the new solutions generated in each iteration are subjected to a non-dominated ranking in order to select the solutions with a high non-dominated rank to proceed to the next round of iterations. Hypervolume (HV) is

an important measure of the size of the region covered by the solution set, and its rate of convergence can reflect the rate of convergence of the solution set of the Pareto front.

Figure 3b demonstrates the evolution of the hypervolume of the three algorithms during the iteration process. From the figure, it can be seen that the proposed QALNS-NSGA-II always has the highest hypervolume of the explored Pareto front and the fastest convergence rate, which means that compared to the other algorithms, QALNS-NSGA-II also has a significant advantage in terms of the convergence rate of the overall solution set.

4.3.2. Convergence Quality Evaluation

To further evaluate the performance of the three algorithms, we conducted a comprehensive comparative analysis of the three algorithms based on the performance indicators defined before. Tables 3–5, list the comparison results of the three algorithms in terms of CM (coverage metric), HV (hypervolume), and SM (spacing matrix). The bold values represent the results of the three algorithms that perform optimally in this indicator. Hit rate records the number of times the algorithm performed best in all cases.

Table 3 demonstrates the coverage metric between several algorithms, and for simplicity, each algorithm is denoted by its initial letter (i.e., CM(Q, A) for QALNS-NSGA-II and ALNS-NSGA-II). From the results, it can be seen that the proposed QALNS-NSGA-II has the highest coverage in all cases, and ALNS-NSGA-II also has an overwhelming advantage over the traditional NSGA-II algorithm. This indicates that compared to the ACO algorithm, the ALNS heuristic operator designed for this problem is more targeted, avoids blind searching, and can converge to a better Pareto frontier solution more efficiently. In addition, using DQN to drive the ALNS heuristic can make full use of other information in the search process; thus, it is less likely to fall into the local optimal solution and has better solving ability and convergence than the traditional heuristic algorithm.

Table 3. The comparison results based on CM.

Case	CM(Q,A)		CM(A,Q)		CM(Q,N)		CM(N,Q)		CM(A,N)		CM(N,A)	
	Mean	**std**	**Mean**	**std**	**Mean**	**std**	**Mean**	**std**	**Mean**	**std**	**Mean**	**std**
SiouxFalls-24-76	**0.79**	0.01	0.12	0.01	**0.86**	0.07	0.02	0.00	**0.71**	0.17	0.18	0.07
gdb-12-44	**0.80**	0.04	0.13	0.05	**0.92**	0.03	0.07	0.03	**0.78**	0.04	0.09	0.04
gdb-22-90	**0.73**	0.13	0.17	0.07	**1.00**	0.00	0.00	0.00	**1.00**	0.00	0.00	0.00
gdb-7-42	**0.87**	0.09	0.12	0.07	**1.00**	0.00	0.00	0.00	**0.63**	0.17	0.14	0.10
val-24-78	**0.85**	0.05	0.06	0.01	**0.93**	0.02	0.00	0.00	**0.98**	0.00	0.00	0.00
val-24-68	**0.68**	0.09	0.14	0.03	**0.98**	0.00	0.01	0.00	**0.91**	0.02	0.08	0.01
val-24-70	**0.65**	0.10	0.03	0.01	**0.83**	0.06	0.08	0.03	**0.54**	0.06	0.27	0.05
kshs-8-30	**0.95**	0.01	0.00	0.00	**0.80**	0.20	0.15	0.11	**0.60**	0.30	0.38	0.27
kshs-10-30	**0.68**	0.11	0.19	0.07	**0.85**	0.03	0.05	0.00	**0.55**	0.03	0.34	0.02
kshs-6-30	**0.97**	0.01	0.03	0.00	**0.87**	0.09	0.03	0.00	**0.58**	0.17	0.36	0.18
Hit rate	**10/10**		0/10		**10/10**		0/10		**10/10**		0/10	

The bold values represent the results of the three algorithms that perform optimally in this indicator.

Table 4 summarizes the results of the HV comparison of the three algorithms. HV denotes the region enclosed by the set of all non-dominated solutions obtained by the algorithms with a reference point in the target space, and a higher value of HV indicates better performance. It can be seen that QALNS-NSGA-II exhibits higher HV values in all the test cases compared to the other algorithms, which further confirms that the QALNS-NSGA-II Pareto frontier solution is more diverse and convergent.

The results of the SM comparison of the three algorithms are summarized in Table 5. It can be seen that QALNS-NSGA-II outperforms the other algorithms in finding diversity solutions in most cases (6/10).

In order to visually demonstrate the effectiveness of the QALNS-NSGA-II algorithm in solving MOP, Figure 4 shows the distribution of the best Pareto front solution sets obtained by three algorithms across five experiments for the nine test cases presented in

this paper. From the figure, it can be observed that the Pareto frontier solutions obtained by the QALNS-NSGA-II algorithm proposed in this paper are generally better than the other compared algorithms in each objective value. This verifies the superior performance of QALNS-NSGA-II in terms of convergence and diversity, which is consistent with the experimental results discussed earlier.

Table 4. The comparison results based on HV.

Case	QALNS-NSGA-II		ALNS-NSGA-II		NSGA-II	
	Mean	**std**	**Mean**	**std**	**Mean**	**std**
SiouxFalls-24-76	**0.89**	0.00	0.80	0.01	0.63	0.04
gdb-12-44	**0.96**	0.00	0.85	0.04	0.74	0.06
gdb-22-90	**0.89**	0.01	0.75	0.02	0.17	0.02
gdb-7-42	**0.87**	0.01	0.67	0.04	0.17	0.05
val-24-78	**0.96**	0.00	0.77	0.01	0.45	0.03
val-24-68	**0.92**	0.01	0.86	0.01	0.60	0.02
val-24-70	**0.86**	0.02	0.79	0.01	0.77	0.02
kshs-8-30	**0.91**	0.01	0.76	0.03	0.69	0.07
kshs-10-30	**0.93**	0.00	0.86	0.00	0.85	0.00
kshs-6-30	**0.98**	0.00	0.76	0.01	0.72	0.03
Hit rate	**10/10**		0/10		0/10	

The bold values represent the results of the three algorithms that perform optimally in this indicator.

Table 5. The comparison results based on SM.

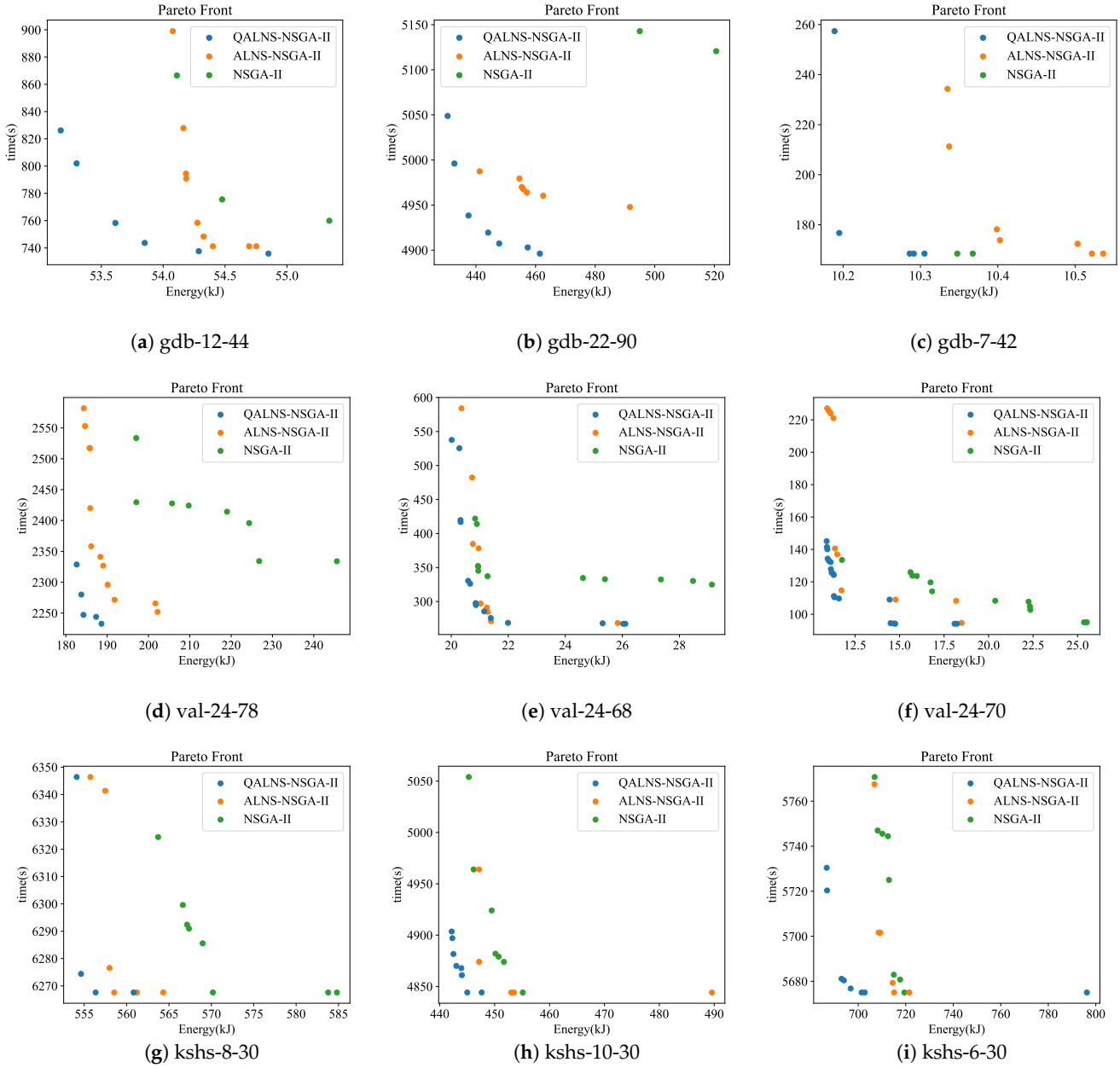
Case	QALNS-NSGA-II		ALNS-NSGA-II		NSGA-II	
	Mean	**std**	**Mean**	**std**	**Mean**	**std**
SiouxFalls-24-76	**3.17**	9.89	9.99	90.11	8.16	29.27
gdb-12-44	10.44	67.21	11.08	67.77	**4.86**	17.36
gdb-22-90	**7.55**	44.29	14.85	279.42	14.56	276.29
gdb-7-42	3.74	10.82	3.69	10.75	**0.68**	/[a]
val-24-78	**5.22**	34.26	15.93	52.71	9.90	42.22
val-24-68	**7.50**	45.41	10.51	157.65	10.03	200.89
val-24-70	**1.02**	0.20	1.63	0.63	3.89	9.27
kshs-8-30	3.92	0.90	4.09	22.39	**2.27**	4.64
kshs-10-30	**3.04**	1.71	7.79	9.82	8.96	48.94
kshs-6-30	6.18	28.51	**3.18**	3.21	4.44	2.46
Hit rate	**6/10**		1/10		3/10	

[a] '/' When the number of Pareto frontier solutions is less than or equal to 3, there are too few sample points to provide valid SM information. In the case of "gdb-7-42", only one piece of SM information is included, and thus, the standard deviation cannot be calculated. The bold values represent the results of the three algorithms that perform optimally in this indicator.

4.4. Trade-Offs Between the Two Objectives

We further examined the relationship between the two objectives. Take the case of "SiouxFalls-24-76" as an example, Figure 5 shows the trade-off between the two objectives, indicating that there is a conflict in the relationship between E and T in the Pareto solution. Therefore, the decision maker needs to make a trade-off between the energy-saving mode and the efficiency mode based on actual needs and choose a solution that meets the current preferences from the Pareto solution set.

From the trend of T in the figure, it can be seen that the process can be divided into two stages: the rapid descent phase and the slow descent phase. In the rapid descent phase, T decreases quickly with a slight increase in E. This is because the unmanned sweepers effectively avoid the red light period of the traffic signal by reasonably planning the path and quickly adjusting the timing of passing through the heavy-duty section in this phase, thus reducing the waiting time. The strategy in this phase tends to sacrifice a small amount of E in exchange for significant time savings, which enables a significant reduction in T.



(**a**) gdb-12-44 (**b**) gdb-22-90 (**c**) gdb-7-42

(**d**) val-24-78 (**e**) val-24-68 (**f**) val-24-70

(**g**) kshs-8-30 (**h**) kshs-10-30 (**i**) kshs-6-30

Figure 4. The Pareto front obtained by the comparison algorithm.

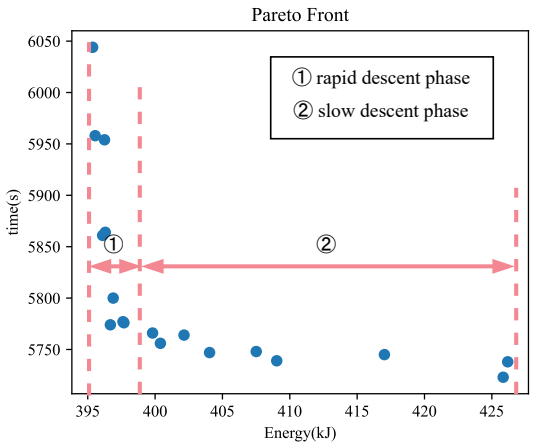


Figure 5. Trade-off between the two objectives in the case "SiouxFalls-24-76".

In the slow descent phase, the rate of increase in E accelerated, but the improvement in T leveled off. This is because at this point, the unmanned sweepers have to avoid traffic lights by detouring, which results in a significant increase in the travel distance. Although the detour strategy avoids the time waste caused by signalization to some extent, it also imposes a high cost of E. Therefore, the optimization space between E and T is limited in this phase, and the improvement in T is weakened.

5. Conclusions

In this paper, we study an unmanned sweeper arc route planning problem with both efficiency- and energy-saving-related objectives (travel time T and energy consumption E) and establish a new multi-objective mixed-integer nonlinear planning model, which takes into account the effects of road slope, dynamic changes in on-board mass, mode switching of vehicle operation, and traffic lights, which makes the model more realistic.

To solve the problem, we design encoding and decoding methods for the arc path planning problem and repair and destruction operators for the adaptive large-scale neighborhood search algorithm based on the features of the problem. In addition, a selected hyperheuristic based on deep reinforcement learning (QALNS) is designed to adaptively select operators during the iteration process using DQN-driven ALNS, and a specialized reward function is designed for the process of Pareto frontier solution iteration to drive accelerated convergence.

QALNS is embedded into non-dominated sorting genetic algorithm II (NSGA-II) to form a new algorithm, QALNS-NSGA-II. We conducted a comprehensive set of experiments to evaluate the performance of our proposed algorithm. We compared QALNS-NSGA-II with the classical NSGA-II and ALNS-NSGA-II. The experimental results show that the proposed algorithm has a remarkable advantage for this problem, both in terms of the speed of exploring Pareto frontier solutions and the quality (diversity and convergence) of the explored Pareto frontier solutions.

In addition, some planning insights can be gained from the experimental analysis. As E increases, T first declines rapidly followed by a gentle decline. The planner can trade-off between these two objectives to find a compromise point suitable for practical needs. In addition, according to the experimental results of unmanned sweeper path planning, the sweeping mode of "light load first, heavy load later" can effectively reduce the unnecessary waste of resources. Under the condition that the total path length remains unchanged, this mode significantly reduces the total trip energy consumption of the unmanned sweeper. Therefore, it is recommended that the initial position of the unmanned sweeper be assigned to neighborhoods with less garbage distribution, to optimize whole path planning and further improve energy utilization efficiency.

The limitations of this study and future research directions are summarized as follows: In this paper, a reinforcement learning integrated metaheuristic algorithm is used to implement adaptive operator selection, and a random initial population and fixed parameters are used in the training process. However, this approach limits the convergence speed and optimization effect of the algorithm to some extent, which may affect its applicability to complex problems. To address this limitation, future research can further optimize the population initialization strategy and explore the dynamic parameter adjustment method to improve the search efficiency and global convergence ability of the algorithm. In addition, future research should also thoroughly analyze the effects of dynamics and stochasticity on path planning in real-world scenarios, combine the advantages of reinforcement learning in high-dimensional data processing, and develop more robust and efficient path planning algorithms for unmanned sweeper vehicles, so as to better meet the needs of practical applications.

Author Contributions: Conceptualization, Y.H. and W.M.; methodology, Y.H.; software, Y.H.; validation, Y.H., W.M. and J.L.; formal analysis, J.L.; investigation, F.L.; resources, K.W.; data curation, W.M.; writing—original draft preparation, Y.H.; writing—review and editing, Y.H.; visualization, Y.H.; supervision, S.L.; project administration, S.L.; funding acquisition, S.L. All authors have read and agreed to the published version of the manuscript.

Funding: This work was financially supported in part by the National Natural Science Foundation of China (Grant No. 52472343, 52302380) and in part by the National Science Foundation of Guangdong Province, China (Grant No. 2023A1515012949).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data used in this study are available from the corresponding author upon reasonable request.

Acknowledgments: The authors are grateful for the hard work of all of the anonymous reviewers to improve the quality of this submission.

Conflicts of Interest: Author Fuhai Luo was employed by the company Fulongma Group Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Nomenclature

Abbreviations

The following abbreviations are used in this manuscript:

SPaT	Traffic Signal, Phasing, and Timing information
OBL	Opposition-based learning
MOP	Multi-objective optimization problem
POF	Pareto Optimal Front
ALNS	Adaptive Large Neighborhood Search
ARP	Arc route Problem
ETM-ARP	Energy and Time Minimization Unmanned Sweeper Arc Route Problem
MINLP	Mixed-Integer Nonlinear Programming
NSGA-II	Non-dominated sorting genetic algorithm II
DQN	Deep Q-Network
HV	Hypervolume
SM	Spacing matrix
CM	Coverage metric

Parameterization

The following parameters are used in this manuscript:

$\delta^+(i)$	The set of nodes toward which there is an arc from node i.
$\delta^-(i)$	The set of nodes from which there is an arc toward node i.
$m_{vehicle}$	The mass of the unmanned sweeper [kg].
$m_{i,j}$	The mass of garbage on the (i,j) road segment [kg].
d_k	The distance of the kth road segment [s].
g	Gravitational acceleration [m/s^2].
A	Frontal surface area [m^2].
ρ	Atmospheric density [kg/m^3].
c_r	Coefficient rolling resistance.
c_d	Coefficient aerodynamic drag.
v_k	Velocity at the kth route segment [km/h].
$\alpha_{i,j,k}$	The binary variable is 1 if the sweeping mode is selected when the unmanned sweeper passes through the (i,j) section for the kth time; otherwise, it is 0.
eff_m^d	Motor drive efficiency [%].
eff_m^r	Motor regenerative braking efficiency [%].
eff_b^d	Battery discharge efficiency [%].
eff_b^r	Battery charging efficiency [%].
$E_{i,j}$	Energy consumption (+) or regenerative braking energy ($-$) on the (i,j) road segment [kJ].
E_k	Accumulated energy consumption up to the kth route segment [kJ].
t_k	Travel time of the kth route segment [s].
M_k	Total mass of unmanned sweeper of the kth route segment, which is defined as the vehicle mass plus the load mass [kg].

k	Route index.
t_{wait}	Wait times for unmanned sweepers at signalized intersections [s].
$\beta_{i,j,k}$	The binary variable is 1 if the traveling mode is selected when the unmanned sweeper passes through the (i, j) section for the kth time; otherwise, it is 0.

References

1. Jeon, J.; Jung, B.; Koo, J.C.; Choi, H.R.; Moon, H.; Pintado, A.; Oh, P. Autonomous Robotic Street Sweeping: Initial Attempt for Curbside Sweeping. In Proceedings of the 2017 IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 8–10 January 2017; pp. 72–73. [CrossRef]
2. Song, B.; Chen, W.; Chen, T.; Zhou, X.; Liu, B. Path Planning in Urban Environment Based on Traffic Condition Perception and Traffic Light Status. In Proceedings of the 2022 IEEE International Symposium on Product Compliance Engineering—Asia (ISPCE-ASIA), Guangzhou, China, 4–6 November 2022; pp. 1–6. [CrossRef]
3. Karoui, M.; Chalhoub, G.; Freitas, A. An Efficient Path Planning GLOSA-based Approach over Large Scale and Realistic Traffic Scenario. *Internet Technol. Lett.* **2021**, *4*, e194. [CrossRef]
4. Wu, L.; Ci, Y.; Chu, J.; Zhang, H. The Influence of Intersections on Fuel Consumption in Urban Arterial Road Traffic: A Single Vehicle Test in Harbin, China. *PLoS ONE* **2015**, *10*, e0137477. [CrossRef] [PubMed]
5. Georgiadis, D.; Karathanasopoulou, K.; Bardaki, C.; Panagiotopoulos, I.; Vondikakis, I.; Paktitis, T.; Dimitrakopoulos, G. Performance Analysis of Energy-Efficient Path Planning for Sustainable Transportation. *Sustainability* **2024**, *16*, 4963. [CrossRef]
6. He, L.; Wang, B.; Peng, Y.; Zhang, X. An Unmanned Sweeper Path Planning Algorithm for Structured Roads. *IEEE Access* **2024**, *12*, 26242–26250. [CrossRef]
7. Ghiani, G.; Guerriero, F.; Improta, G.; Musmanno, R. Waste Collection in Southern Italy: Solution of a Real-Life Arc Routing Problem. *Int. Trans. Oper. Res.* **2005**, *12*, 135–144. [CrossRef]
8. Blazquez, C.A.; Beghelli, A.; Meneses, V.P. A Novel Methodology for Determining Low-Cost Fine Particulate Matter Street Sweeping Routes. *J. Air Waste Manag. Assoc.* **2012**, *62*, 242–251. [CrossRef]
9. Song, M.; Lu, B.; Cheng, L.; Sun, C. Lagrangian Relaxation-Based Decomposition Approaches for the Capacitated Arc Routing Problem in the State-Space-Time Network. *Transp. Lett.* **2023**, *15*, 1317–1336. [CrossRef]
10. Lu, J.; Nie, Q.; Mahmoudi, M.; Ou, J.; Li, C.; Zhou, X.S. Rich Arc Routing Problem in City Logistics: Models and Solution Algorithms Using a Fluid Queue-Based Time-Dependent Travel Time Representation. *Transp. Res. Part B: Methodol.* **2022**, *166*, 143–182. [CrossRef]
11. Huang, Z.; Zheng, Q.P.; Pasiliao, E.; Boginski, V.; Zhang, T. A Cutting Plane Method for Risk-Constrained Traveling Salesman Problem with Random Arc Costs. *J. Glob. Optim.* **2019**, *74*, 839–859. [CrossRef]
12. Ku, D.; Choi, M.; Yoo, N.; Shin, S.; Lee, S. A New Algorithm for Eco-Friendly Path Guidance Focused on Electric Vehicles. *Energy* **2021**, *233*, 121198. [CrossRef]
13. Liu, C.; Wang, J.; Cai, W.; Zhang, Y. An Energy-Efficient Dynamic Route Optimization Algorithm for Connected and Automated Vehicles Using Velocity-Space-Time Networks. *IEEE Access* **2019**, *7*, 108866–108877. [CrossRef]
14. Liu, G.; Zhao, H.; Yang, H.; Zhiwei, D.W.; Wang, D. A Fleet Management System of Autonomous Electric Street Sweepers. In Proceedings of the 2023 IEEE International Conference on Cybernetics and Intelligent Systems (CIS) and IEEE Conference on Robotics, Automation and Mechatronics (RAM), Penang, Malaysia, 9–12 June 2023; pp. 152–157. [CrossRef]
15. Perger, T.; Auer, H. Energy Efficient Route Planning for Electric Vehicles with Special Consideration of the Topography and Battery Lifetime. *Energy Effic.* **2020**, *13*, 1705–1726. [CrossRef]
16. Boriboonsomsin, K.; Barth, M.J.; Zhu, W.; Vu, A. Eco-routing navigation system based on multisource historical and real-time traffic information. *IEEE Trans. Intell. Transp. Syst.* **2012**, *13*, 1694–1704. [CrossRef]
17. Golden, B.L.; Wong, R.T. Capacitated Arc Routing Problems. *Networks* **1981**, *11*, 305–315. [CrossRef]
18. Hu, J.; Luo, X.; Gan, Y. Robust electric vehicle routing problem based on column generation algorithm[in Chinese]. *Comput. Integr. Manuf. Syst.* **2023**, *29*, 2427–2439. [CrossRef]
19. Nie, Q.; Long, X.; Liang, C.; Xia, J.; Ou, J. Optimal Allocation and Routing of Sanitation Vehicles on Urban Roads Under Multiple Constraints. *J. Transp. Syst. Eng. Inf. Technol.* **2022**, *22*, 273–284. (In Chinese) [CrossRef]
20. Huang, S.H.; Lin, T.H. Using Ant Colony Optimization to Solve Periodic Arc Routing Problem with Refill Points. *J. Ind. Prod. Eng.* **2014**, *31*, 441–451. [CrossRef]
21. Tirkolaee, E.; Hosseinabadi, A.; Soltani, M.; Sangaiah, A.; Wang, J. A Hybrid Genetic Algorithm for Multi-Trip Green Capacitated Arc Routing Problem in the Scope of Urban Services. *Sustainability* **2018**, *10*, 1366. [CrossRef]
22. Khajepour, A.; Sheikhmohammady, M.; Nikbakhsh, E. Field Path Planning Using Capacitated Arc Routing Problem. *Comput. Electron. Agric.* **2020**, *173*, 105401. [CrossRef]
23. Sutton, R.; Barto, A. *Reinforcement Learning: An Introduction*; A Bradford Book: Cambridge, MA, USA, 2018.
24. Kallestad, J.; Hasibi, R.; Hemmati, A.; Sörensen, K. A General Deep Reinforcement Learning Hyperheuristic Framework for Solving Combinatorial Optimization Problems. *Eur. J. Oper. Res.* **2023**, *309*, 446–468. [CrossRef]
25. Durgut, R.; Aydin, M.E.; Atli, I. Adaptive Operator Selection with Reinforcement Learning. *Inf. Sci.* **2021**, *581*, 773–790. [CrossRef]
26. Li, P.; Xue, Q.; Zhang, Z.; Chen, J.; Zhou, D. Multi-Objective Energy-Efficient Hybrid Flow Shop Scheduling Using Q-learning and GVNS Driven NSGA-II. *Comput. Oper. Res.* **2023**, *159*, 106360. [CrossRef]

27. Luo, W.; Yu, X. Reinforcement learning-based modified cuckoo search algorithm for economic dispatch problems. *Knowl.-Based Syst.* **2022**, *257*, 109844. [CrossRef]
28. Kalyanmoy, D. A fast and elitist multi-objective genetic algorithm: NSGA-II. *IEEE Trans. Evol. Comput.* **2002**, *6*, 182–197.
29. Snail White Smart Cleaning Leading Brand Official Website. 2023. Available online: https://www.idriverplus.com/cn/index.html (accessed on 18 September 2024).
30. Belenguer, J.M.; Benavent, E. A Cutting Plane Algorithm for the Capacitated Arc Routing Problem. *Comput. Oper. Res.* **2003**, *30*, 705–728. [CrossRef]
31. Benavent, E.; Campos, V.; Corberan, A.; Mota, E. The Capacitated Arc Routing Problem: Lower Bounds. *Networks* **1992**, *22*, 669–690. [CrossRef]
32. Golden, B.L.; DeArmon, J.S.; Baker, E.K. Computational experiments with algorithms for a class of routing problems. *Comput. Oper. Res.* **1983**, *10*, 47–59. [CrossRef]
33. Kiuchi, M.; Shinano, Y.; Hirabayashi, R.; Saruwatari, Y. An exact algorithm for the capacitated arc routing problem using parallel branch and bound method. In Proceedings of the Spring National Conference of the Operational Research Society of Japan, INFORMS, Catonsville, MD, USA, 15 June 1995; pp. 28–29.
34. Transportation Networks for Research. 2023. Available online: https://github.com/bstabler/TransportationNetworks (accessed on 18 September 2024).
35. Ding, J.Y.; Song, S.; Wu, C. Carbon-efficient scheduling of flow shops by multi-objective optimization. *Eur. J. Oper. Res.* **2016**, *248*, 758–771. [CrossRef]
36. Zitzler, E.; Thiele, L. Multiobjective Evolutionary Algorithms: A Comparative Case Study and the Strength Pareto Approach. *IEEE Trans. Evol. Comput.* **1999**, *3*, 257–271. [CrossRef]
37. Wang, H.; Fu, Y.; Huang, M.; Huang, G.Q.; Wang, J. A NSGA-II based memetic algorithm for multiobjective parallel flowshop scheduling problem. *Comput. Ind. Eng.* **2017**, *113*, 185–194. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.